# DIETARY ELIMINATION OF Neu5Gc, GLIADINS, AND AMYLASE-TRYPSIN INHIBITORS AS A STRATEGY TO REDUCE CHRONIC INFLAMMATORY LOAD AND POTENTIATE IMMUNE-MEDIATED CANCER SURVEILLANCE

*A Hypothesis Linking Dietary Xenoantigen Elimination to Tumour Microenvironment Modification, Immune Reconstitution, and Synergistic Enhancement of Glycan-Targeting Biotherapeutics*

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## Abstract

Background: Chronic low-grade inflammation is a well-established driver of tumour initiation, progression, and immune evasion. Cancer cells exploit the sialic acid-rich glycocalyx as a molecular shield, delivering inhibitory signals to immune effector cells and physically obstructing cytotoxic engagement. Two dietary sources of persistent systemic inflammation have received insufficient attention in oncological nutritional research: dietary N-glycolylneuraminic acid (Neu5Gc) from red meat and bovine dairy, which is incorporated into human tissues and elicits a chronic antibody-mediated inflammatory response termed xenosialitis; and wheat-derived gliadins and amylase-trypsin inhibitors (ATIs), which activate innate immune pathways and compromise intestinal barrier integrity, producing the systemic immune drain of metabolic endotoxaemia.

Hypothesis: We propose that concurrent dietary elimination of Neu5Gc, gliadins, and ATIs substantially reduces the systemic inflammatory burden in susceptible individuals including cancer patients, liberating immune resources and reconstituting robust, targeted anti-tumour surveillance. We further propose that even when cancer cells adapt by rebuilding their glycocalyx shield using the native human sialic acid Neu5Ac in place of Neu5Gc, the newly unburdened immune system is significantly better positioned to act synergistically with sialidase-based or lectin-hybrid biotherapeutics to expose and destroy shielded tumour cells. This dietary modification is not proposed as curative monotherapy — it is framed explicitly as an essential adjunct that restructures the immune landscape, removes tumour-specific growth stimuli, and primes the immune system for maximal efficacy of emerging glycan-targeting therapies.

Significance: Cancer immunology has increasingly recognised the glycocalyx as a target for therapeutic intervention. The present hypothesis adds a dimension that immunological research has not previously examined: the role of dietary Neu5Gc in shaping the glycocalyx composition and inflammatory microenvironment of tumours, and the potential for dietary elimination to shift that microenvironment in favour of immune-mediated tumour destruction.

**Keywords:** *Neu5Gc; sialic acid; glycocalyx; xenosialitis; gliadins; amylase-trypsin inhibitors; ATI; cancer immunology; immune exhaustion; Neu5Ac; sialidase; glycan shield; tumour microenvironment; dietary inflammation; immune reconstitution; biotherapeutics*

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## 1. Introduction: The Glycocalyx Problem and the Dietary Dimension

Cancer cells are distinguished by their capacity to evade immune destruction through multiple overlapping mechanisms. Among the most structurally sophisticated is the assembly of a dense, sialic acid-rich glycocalyx — a sugar-coated extracellular shield that physically obstructs immune effector cell engagement and delivers inhibitory 'don't eat me' signals to macrophages and natural killer (NK) cells via Siglec receptors. This glycocalyx shield is not static; it is dynamically regulated by the tumour in response to its immunological environment, and its composition — specifically the ratio of Neu5Gc to Neu5Ac in its sialic acid complement — is substantially shaped by the patient's dietary environment.

What has received comparatively limited attention in mainstream oncological literature is the degree to which that dietary environment contributes not only to glycocalyx composition but to the systemic immune state within which tumour cells operate. A chronically inflamed, immunologically exhausted host is a fundamentally different therapeutic environment than a host whose immune system has been reconstituted and whose anti-tumour surveillance is operating at full capacity. The question of how to achieve that reconstitution has focused almost exclusively on pharmacological immunomodulation — checkpoint inhibitors, CAR-T cell therapy, bispecific antibodies. The dietary dimension of immune reconstitution has received a fraction of the attention it warrants.

This paper presents a formal hypothesis that the concurrent elimination of two specific dietary xenoantigens — Neu5Gc and wheat-derived ATIs and gliadins — will: deprive tumours of a key glycocalyx building block and its associated inflammatory growth stimuli; reduce systemic immune exhaustion and reconstitute robust, targeted anti-tumour immunity; and create a biologically optimised immune environment in which sialic acid-targeting biotherapeutics can operate at peak efficacy. It is offered not as a challenge to current oncological management but as a mechanistically grounded complement to it.

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## 2. Background

### 2.1 Neu5Gc as a Dietary Xenoantigen and Tumour Growth Factor

Humans are genetically unique among great apes in their inability to synthesise Neu5Gc, due to an inactivating mutation in the CMAH gene approximately 2–3 million years ago. Despite this, dietary consumption of Neu5Gc — present in red meat and bovine dairy at concentrations of 5–30 mg/kg and 7–11 mg/L respectively — results in passive absorption and incorporation of this foreign sialic acid into human cell-surface glycans, a process confirmed in human-like CMAH-knockout mouse models and in direct human tissue analysis.

Because Neu5Gc is immunologically foreign, the human immune system generates circulating anti-Neu5Gc antibodies. When these antibodies encounter Neu5Gc-decorated tissues — particularly within tumours, which appear to preferentially sequester dietary Neu5Gc — they trigger a low-level, self-sustaining inflammatory response termed xenosialitis. This response is, paradoxically, tumour-promoting rather than tumour-suppressive: at the low amplitudes characteristic of chronic dietary exposure, anti-Neu5Gc immune complex formation stimulates angiogenesis through VEGF upregulation, activates proliferative signalling cascades including the Wnt/β-catenin pathway, and creates a tumour microenvironment that favours growth and immune evasion rather than immune attack. The very antibody response that should be fighting the foreign sialic acid is instead providing the tumour with growth signals.

Samraj and colleagues demonstrated this mechanism directly: feeding a Neu5Gc-rich diet to CMAH-knockout mice — the animal model that most closely approximates the human scenario of dietary Neu5Gc in the absence of endogenous synthesis — significantly accelerated tumour growth and promoted cancer progression compared to control diets, through a mechanism involving chronic xenosialitis-driven tumour microenvironment inflammation rather than direct carcinogenesis.

## 2.2 Gliadins and ATIs as Drivers of Systemic Immune Drain

Gliadins are storage proteins in wheat gluten that, in susceptible individuals, breach intestinal epithelial integrity through zonulin-mediated tight junction disruption and translocate into systemic circulation. Amylase-trypsin inhibitors (ATIs), also derived from modern wheat, are potent activators of TLR4 on innate immune cells, triggering release of TNF-α, IL-1β, IL-6, and IL-8 in gut-associated and systemic immune compartments independently of celiac autoimmunity.

The resulting chronic, low-grade systemic inflammation creates what this framework terms an immune drain: a persistent, energetically costly, non-targeted immune activation that consumes white blood cell reserves, inflammatory mediators, and regulatory capacity that would otherwise be available for directed anti-tumour surveillance. An immune system perpetually engaged with dietary antigens is an immune system with reduced resources available for the recognition and destruction of malignant cells. The wheat-derived component of the dietary trigger burden is not a minor cofactor in this framework — it is a major driver of the systemic immune exhaustion that allows tumours to escape detection and destruction.

## 2.3 The Adaptive Glycocalyx: The Neu5Ac Substitution Challenge

A critical caveat to any dietary Neu5Gc elimination strategy must be addressed directly: cancer cells demonstrate metabolic plasticity in their glycocalyx construction. When deprived of dietary Neu5Gc, tumours do not lose their glycan shield. Instead, they upregulate incorporation of Neu5Ac — the native human sialic acid — to reconstruct a functionally analogous protective coat. This substitution preserves the tumour's capacity for immune evasion through Siglec-mediated inhibitory signalling and physical obstruction of immune effector engagement.

This adaptive response represents the central paradox that this hypothesis resolves. Dietary Neu5Gc elimination does not destroy the glycocalyx shield. It changes its composition — from a Neu5Gc-bearing shield that simultaneously evades immunity and provides inflammatory growth signals, to a Neu5Ac-bearing shield that evades immunity but no longer provides those growth signals. And a Neu5Ac-bearing shield, unlike a Neu5Gc-bearing one, is precisely the target for which the most promising sialic acid-stripping biotherapeutics have been designed. The dietary intervention and the biotherapeutic are not competing strategies; they are sequential and mutually potentiating ones.

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## 3. Formal Hypothesis

We propose the following testable hypothesis:

> *In cancer patients, the concurrent dietary elimination of Neu5Gc (via avoidance of red meat and bovine dairy) and gliadins/ATIs (via avoidance of wheat and related grains) will: (1) extinguish xenosialitis-driven tumour angiogenesis and Wnt/β-catenin proliferative signalling; (2) reduce systemic immune exhaustion and reconstitute robust, targeted NK cell and cytotoxic T-cell anti-tumour activity; and (3) create an immunologically primed environment in which — even after tumours adapt by substituting Neu5Ac for Neu5Gc in the glycocalyx — the combination of a strengthened immune system and glycan-targeting biotherapeutics is substantially more effective at exposing and destroying shielded tumour cells than in the chronic inflammatory baseline state.*

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## 4. Mechanistic Rationale

### 4.1 Withdrawal of Tumour Fertilizer: Extinguishing Xenosialitis

The xenosialitis cycle is self-reinforcing while dietary Neu5Gc supply persists. Anti-Neu5Gc antibodies encounter Neu5Gc in the tumour microenvironment, triggering complement activation and cytokine release that, at the low levels characteristic of chronic dietary xenosialitis, promotes rather than resolves tumour growth. Angiogenic signals supply the tumour with increasing vascular access; Wnt/β-catenin activation drives cellular proliferation and resistance to apoptosis.

Dietary elimination of Neu5Gc breaks this cycle at its source. As circulating dietary Neu5Gc is progressively cleared from tissues over weeks to months following elimination, the antigenic stimulus for xenosialitis diminishes and the tumour loses its inflammatory growth advantage. This does not destroy the tumour — it fundamentally alters the tumour microenvironment from a pro-growth state to a growth-neutral state with respect to this specific pathway, removing a chronic fuel source from a fire that standard oncological management is simultaneously attempting to extinguish from other directions.

## 4.2 Immune Reconstitution: Liberating Anti-Tumour Effector Capacity

Concurrent elimination of gliadins and ATIs halts TLR4-mediated innate immune activation and reduces intestinal permeability, substantially lowering the systemic cytokine background and ending the immune drain that wheat-derived antigens maintain. The immune system, freed from the metabolic burden of sustained diffuse inflammation directed at dietary antigens, can reallocate resources toward organised anti-tumour surveillance.

Specifically, dietary elimination is predicted to produce the following immune reconstitution effects:

| Immune Component | Effect of Chronic Dietary Antigen Load | Predicted Effect of Elimination |
|---|---|---|
| NK cell cytotoxicity | Depressed by systemic inflammatory exhaustion; reduced capacity to patrol for and destroy tumour cells with reduced MHC-I expression | Restored cytotoxic capacity; increased tumour cell killing in ex vivo assays |
| Cytotoxic T-lymphocytes (CTL) | Reduced infiltration into tumour microenvironment; increased regulatory T-cell suppression from chronic inflammatory state | Reduced Treg suppression; more robust CTL infiltration into tumour territory |
| Tumour-associated macrophages | Polarised toward pro-tumorigenic M2 phenotype by chronic IL-4/IL-13 and TGF-β signalling | Shift toward tumour-suppressive M1 phenotype as systemic inflammatory drivers diminish |
| Systemic cytokine milieu | Elevated TNF-α, IL-1β, IL-6, IL-8 creating inflammatory noise that obscures tumour-specific immune signalling | Normalisation of baseline cytokines; cleaner immunological signal allowing tumour-specific recognition to dominate |

*Table 1. Predicted immune reconstitution effects of concurrent dietary Neu5Gc and ATI/gliadin elimination in cancer patients.*

## 4.3 Resolving the Neu5Ac Substitution Paradox

When cancer cells substitute Neu5Ac for Neu5Gc in the glycocalyx following dietary Neu5Gc elimination, they regain a functional immune shield — but one with a fundamentally different immunological profile that is actually more therapeutically tractable:

- Neu5Ac is a self-antigen. The immune system does not mount antibody responses against it, xenosialitis does not occur at the tumour surface, and the tumour permanently loses its inflammatory growth advantage regardless of how efficiently it reconstructs the shield

- Neu5Ac-bearing shields remain precisely targetable. Sialidase enzymes cleave Neu5Ac from the tumour surface with the same precision as Neu5Gc; AbLec hybrid molecules bind and neutralise sialic acid residues regardless of their specific molecular identity
- A reconstituted immune system amplifies the therapeutic window. When sialidase or AbLec therapy strips the Neu5Ac shield, the exposed tumour cell encounters an immune system that is no longer exhausted by dietary antigen burden — one with fully operational NK cells, cytotoxic T-cells, and tumour-activated macrophages primed for immediate cytotoxic response

The net result is a biologically synergistic system in which each component potentiates the others: dietary modification removes the tumour's inflammatory fuel and restores immune fitness; glycan-targeting biotherapy dismantles the rebuilt shield; and a robust, reconstituted immune system completes the cytotoxic response. None of these components alone achieves what their sequential combination can.

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## 5. The Cascade: From Dietary Antigen to Tumour Microenvironment

| Stage | Event | Tumour Consequence | Intervention Point |
|---|---|---|---|
| 1 | Dietary Neu5Gc ingested; absorbed; incorporated into tumour glycoproteins | Tumour glycocalyx enriched with Neu5Gc; anti-Neu5Gc antibodies recognise tumour surface | Dietary elimination removes supply |
| 2 | Anti-Neu5Gc IgG engages tumour-incorporated Neu5Gc; immune complex formation at tumour surface | Complement activation; local cytokine release — pro-tumorigenic at xenosialitis concentrations | Elimination breaks antigenic stimulus |
| 3 | Xenosialitis-driven cytokine release stimulates tumour angiogenesis (VEGF) and proliferation (Wnt/β-catenin) | Accelerated vascular supply; enhanced tumour growth; increased metastatic capacity | Elimination removes chronic growth stimulus |
| 4 | Systemic immune drain from ATI/gliadin-driven TLR4 activation and LPS translocation | Reduced NK cell and CTL anti-tumour capacity; M2 macrophage polarisation favours tumour progression | Wheat elimination ends immune drain; immune reconstitution begins |
| 5 | Tumour adapts: substitutes Neu5Ac for Neu5Gc in | Glycan shield maintained; immune evasion preserved; but | Sialidase/AbLec biotherapy strips Neu5Ac shield; reconstituted immune system responds |

| Stage | Event | Tumour Consequence | Intervention Point |
|---|---|---|---|
| | glycocalyx following dietary elimination | inflammatory growth signals permanently withdrawn | |
| 6 | Reconstituted immune system engages unshielded tumour cell | Maximum anti-tumour cytotoxicity in absence of both inflammatory noise and glycan shield | Combined dietary + biotherapeutic approach achieves synergistic outcome |

*Table 2. The dietary xenoantigen-to-tumour-microenvironment cascade, with intervention points. The dietary component addresses Stages 1–4; biotherapeutics address Stage 5; the reconstituted immune system executes Stage 6.*

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## 6. Testable Predictions

| Prediction | Investigation | Expected Finding |
|---|---|---|
| Cancer patients will show measurable reductions in circulating xenosialitis and ATI-driven inflammatory markers following dietary elimination | Structured elimination trial in cancer patients (8–16 weeks); primary: serum anti-Neu5Gc IgG, IL-6, TNF-α, IL-8, hsCRP; secondary: NK cell cytotoxicity assay ex vivo | Reduced anti-Neu5Gc titres; reduced systemic inflammatory markers; improved NK cell cytotoxicity |
| Tumour biopsies will show glycocalyx shift from Neu5Gc-dominant to Neu5Ac-dominant following dietary elimination | Paired pre/post biopsy analysis using mass spectrometry sialic acid quantification; concurrent VEGF and Wnt/β-catenin pathway marker analysis in tumour tissue | Reduced tumour Neu5Gc content; reduced VEGF and active β-catenin; shift to Neu5Ac-predominant glycocalyx |
| NK cell cytotoxicity will be higher in dietary-elimination patients than standard-diet controls | Peripheral blood NK cell cytotoxicity assay (ex vivo tumour cell killing) in elimination vs control groups; controlling for tumour type, stage, and treatment regimen | Significantly higher tumour cell killing in elimination group; correlation with reduction in systemic inflammatory markers |
| Cancer patients on elimination diet receiving adjuvant sialidase or AbLec biotherapy will show superior tumour response rates | Randomised controlled trial: biotherapy alone vs dietary elimination + biotherapy; primary: tumour response rate at 12 weeks; secondary: progression-free survival, overall survival, immune cell infiltration on biopsy | Superior tumour response rate in combined arm; NK and CTL infiltration higher in combined arm biopsies |

| Prediction | Investigation | Expected Finding |
|---|---|---|
| Mouse xenograft models supplemented with dietary Neu5Gc and gluten analogues will show accelerated tumour angiogenesis and glycocalyx Neu5Gc density | CMAH-knockout mouse xenograft model: Neu5Gc-supplemented vs control diet; primary: tumour growth rate, VEGF, vessel density, tumour glycocalyx Neu5Gc content | Faster tumour growth; higher VEGF; greater vessel density; higher Neu5Gc content in glycocalyx of Neu5Gc-supplemented group |

*Table 3. Testable Predictions. Prediction 4 is the highest-value clinical test: a randomised trial of biotherapy alone versus dietary elimination plus biotherapy directly tests the synergistic hypothesis that is the paper's central claim.*

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## 7. Limitations and Considerations

Several limitations must be acknowledged explicitly. The timeline for immune reconstitution following dietary elimination is unknown and likely varies by individual immune history, prior treatment exposures, tumour burden, and the degree of immune exhaustion at the time of elimination. Patients with advanced disease and severe immunosuppression from prior chemotherapy may experience less complete reconstitution than treatment-naive patients.

Patient adherence to a strict combined elimination diet — particularly in the context of cancer-associated cachexia, treatment-related nausea, and nutritional vulnerability — presents a significant clinical challenge. The dietary restriction required (no red meat, no bovine dairy, no wheat) is not trivial, and the hidden sources of both Neu5Gc and ATIs in processed foods — documented in Appendix II of the Hamer Loop Manuscript Series — make inadvertent re-exposure a realistic risk without specialised ingredient-identification support.

The degree of Neu5Gc-to-Neu5Ac substitution in different tumour types may vary considerably, affecting the predicted immunological shift and the window of opportunity for biotherapeutic intervention. Some tumour types may substitute Neu5Ac rapidly and completely; others may show incomplete substitution or may upregulate non-sialic acid-based immune evasion mechanisms as compensatory responses.

This hypothesis does not propose that dietary elimination alone is curative. It is framed explicitly as an adjunctive strategy to enhance the efficacy of active oncological therapies. The hypothesis specifically predicts synergy with biotherapeutics; it does not predict meaningful tumour regression from dietary change alone in established malignancy.

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## 8. Conclusion

Cancer immunology has achieved remarkable advances through checkpoint inhibition, CAR-T cell therapy, and the emerging field of glycan-targeting biotherapeutics. What it has not yet fully incorporated is the recognition that the immune system operating in a state of chronic dietary antigen-driven exhaustion is a fundamentally different — and less therapeutically responsive — system than one that has been liberated from that burden.

The tumour does not exist in isolation. It exists within a host whose immune capacity has been shaped, in part, by decades of dietary Neu5Gc and ATI/gliadin exposure. That host's immune system is not failing solely because the tumour is clever; it is failing because it is exhausted, because its resources are perpetually directed at dietary antigens that arrive with every meal, and because the tumour has learned to exploit both the xenosialitis-driven growth signals and the immunological noise that diet creates.

Remove the dietary antigens, and three things happen simultaneously: the tumour loses its inflammatory growth fuel; the immune system recovers its capacity for targeted anti-tumour surveillance; and the glycocalyx composition shifts to a form that is precisely the target for which the most promising glycan-targeting biotherapeutics have been engineered. The dietary intervention does not replace the biotherapeutic. It prepares the battlefield for it.

This is not a fringe idea. It is a mechanistically grounded, experimentally testable hypothesis built on the published work of the world's leading glycobiologists and cancer immunologists. It requires clinical investigation. And the patients who may benefit from it deserve that investigation to begin.

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